

Final Project Phase 2 Report

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Abstract

This report represents a simulated digital forensic investigation conducted on a suspected computer system to locate hidden evidence in the form of Bible Verses. The primary goal of this project is to preserve the integrity of recovered evidence, while attempting to recover relevant data from the device. A systematic methodology was used to investigate the evidence, including recovering hidden files, keyword searching to identify potential evidence, and examining hidden data storage. Moreover, the investigation team created a forensic image of the original drive, using cryptographic hashes to ensure file integrity so that the evidence maintains admissibility in court. Despite these efforts, the investigation resulted in limited results. Six recovered files were identified, however no relevant evidence was found. Difficulties mounting disk images and potential drive corruption restricted further analysis. Overall, the investigation did not produce any verifiable Bible verse evidence, due to the limitation of compromised or inaccessible data.

Introduction

In today's world, cybercrime runs rampant. From ransomware to cyber terrorism, the need for cybersecurity professionals has never been more important. While most people see cybersecurity as defending systems from malicious actors, the investigation of digital evidence is just as vital. Organizations rely on these forensic investigations to respond to incidents, prosecute criminals, and recover critical data. In this project, we simulated a scenario where we acquired a computer with hidden evidence by switching laptops with another team. We received the laptop from the opposing team on Friday, March 27 at 4:15 PM. The laptop is an HP ProBook 640 G5, with serial number 5CG020CCXW. The evidence that we are attempting to find in this scenario are bible verses, and Prof. Daniel Kender gave us the authority to conduct an investigation. In doing so, we went through the process of a forensic investigation to maintain chain of custody and ensure data is not tampered with. This project reflects real-world practices used to ensure that digital evidence remains admissible in court. Overall, the goal of our investigation is to perform forensic analysis techniques to discover evidence left by the users of the computer.

Literature Review

Throughout this project, various tools were used to examine the drive of the laptop we were investigating. A necessary tool that we used in this lab was PhotoRec, a tool used for file carving. In forensics, file carving is the method of recovering files that have been deleted or corrupted (Yoo et. al, 2012). On the drive, the smallest logical unit of disk space is called a cluster. The file system uses these clusters to store files. When a file is deleted on a computer, the file system removes the file's entry from a directory table. The clusters that held the file are marked as unallocated, but the data remains in the cluster until it is overwritten (Pal & Memon, 2009). With a tool such as PhotoRec, we can find this data from the unallocated space.

According to Fikri (2017), PhotoRec works by examining the entire disk sector by sector, identifying file signatures based on file headers. For example, JPG has the header FFD8. Extracted data is then saved into new files, but the original file names and folder structure are not recovered. Overall, the file carving tool PhotoRec was used to recover files from unallocated space.

After mounting the drive, one of the techniques that could be implemented is utilizing Alternate Data Streams (ADS). In the Windows file system, a single file contains a main data stream, which displays what we see in the file, and hidden alternate streams. According to Means (2003), to understand ADS, you have to understand the Master File Table (MFT). The MFT is a special file that stores metadata for every file and folder. Every file or directory has a record in the MFT, containing information such as file name or permissions. The default data attribute stored on the MFT record is normal visible file content, and any additional data attributes are called ADS (Means, 2023). By analyzing ADS of files, an investigator can find data that a suspect might hide.

Another useful tool used in this investigation was the command line utility grep. Magloire (2000) states, “grep searches the input files for lines containing a match to a given pattern list” (p. 1). This tool can search files, multiple files, or standard input piped from other commands. Grep has various capabilities, including flags for case-insensitive search and even recursive directory searching. By using grep paired with other commands, specific phrases can be extracted from files.

Methods

Evidence Acquisition

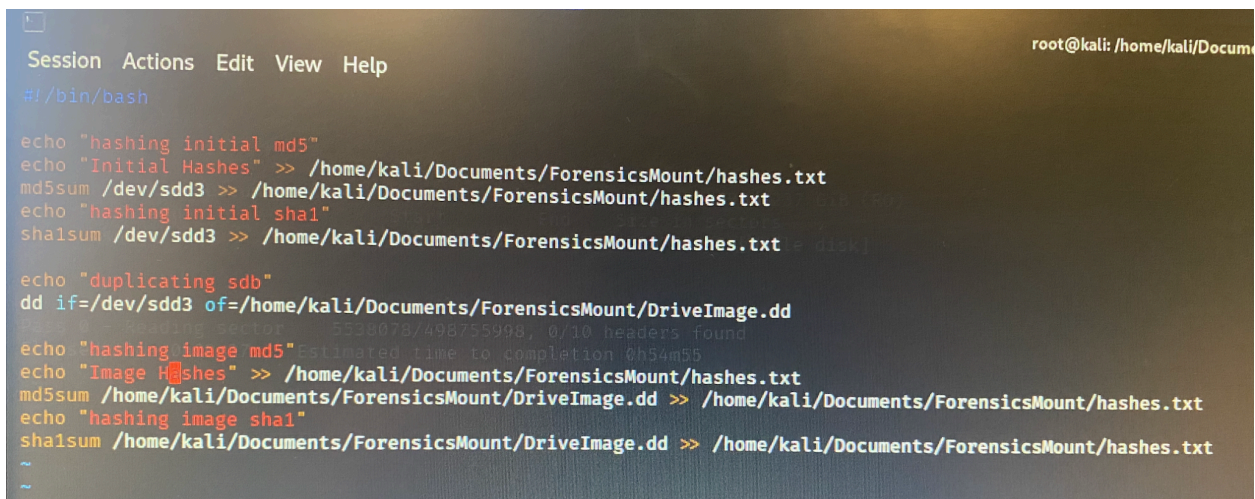
The investigation began by physically disassembling the suspect computer. The internal storage drive and battery pack from the suspect computer were removed. To avoid contaminating the data, we ensured that the proper handling of the evidence was accessible only to the authorized users who were investigating the drive from the suspect.

Initial Disk Imaging Script

An initial script was developed to create a forensic image of the entire drive. Figure 1 displays the script used. A script was created to ensure forensic integrity throughout the imaging process. Hashes were calculated from the original, opposing team's drive. After creating a forensic image of the drive onto a separate analysis disk, hashes were calculated from the image. Figure 2 exhibits the script being executed. However, this approach proved inefficient for analysis due to the size and complexity of the full disk image, and one record was not copied over. This resulted in our hashes being different, because of a single record not imaging correctly. At this point, we were told to continue investigating our image due to time constraints.

Figure 1

Imaging and File Integrity Script. March 30th, 2026 4:46 PM



```
Session Actions Edit View Help
root@kali: /home/kali/Documents/ForensicsMount/

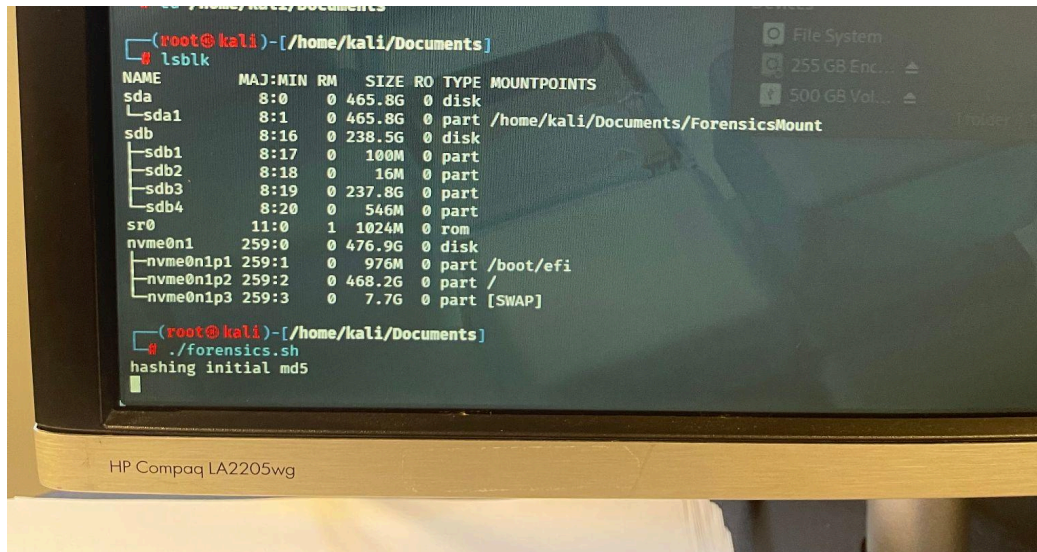
# /bin/bash

echo "hashing initial md5"
echo "Initial Hashes" >> /home/kali/Documents/ForensicsMount/Hashes.txt
md5sum /dev/sdd3 >> /home/kali/Documents/ForensicsMount/Hashes.txt
echo "hashing initial sha1"
sha1sum /dev/sdd3 >> /home/kali/Documents/ForensicsMount/Hashes.txt

echo "duplicating sdb"
dd if=/dev/sdd3 of=/home/kali/Documents/ForensicsMount/DriveImage.dd
dd: writing to: /home/kali/Documents/ForensicsMount/DriveImage.dd: 0/10 headers found
echo "hashing image md5" Estimated time to completion: 2h54m55s
echo "Image Hashes" >> /home/kali/Documents/ForensicsMount/Hashes.txt
md5sum /home/kali/Documents/ForensicsMount/DriveImage.dd >> /home/kali/Documents/ForensicsMount/Hashes.txt
echo "hashing image sha1"
sha1sum /home/kali/Documents/ForensicsMount/DriveImage.dd >> /home/kali/Documents/ForensicsMount/Hashes.txt
~
~
```

Figure 2

Script Execution. March 30th, 2026 4:48 PM

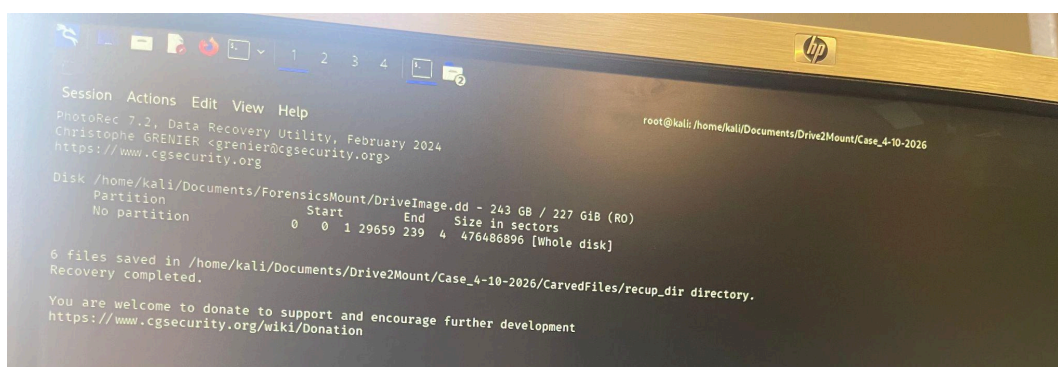


File Carving

File carving was initiated to recover deleted or unallocated files, utilizing the tool PhotoRec. File carving techniques were used to extract data directly from raw disk space. Recovered files were stored on a separate drive to maintain evidence integrity. A total of six files were successfully recovered through this process, and demonstrated in Figure 3.

Figure 3

File Carving Results. April 13th, 2026 3:57 PM



These six files were investigated using the strings command to extract printable character sequences from, and the output was piped to the grep command to locate a pattern based on

regular expressions. This regular expression pattern “[A-Z][A-Za-z]* [1-9]*:[1-9]*”, which would explain bible verses in the form "Book Chapter:Verse". To find any alternate data streams, the command “cat filename:stream name” would be able to output any hidden streams, upon finding the name of the stream. After analyzing these files, all six files all ended up being temporary cache files, which held no Bible verses.

Mounting Issues

Another way to investigate the drive is to mount the image as the drive to investigate the file system. Upon attempting this, we were unable to mount the drive. Attempting to mount the original drive itself also proved ineffective. The possible reasons for this include a bitlocker-encrypted drive, a corrupted drive, or a drive that is not NTFS.

Script Re-Imaging

Our last attempt in investigating the drive was to re-image it. After using the same script, we ran strings command on the image with grep to try and find any Bible verses in plaintext. However, this resulted in no Bible verses being found.

Results

Our only findings from this investigation were the six carved files from PhotoRec, found by Landon and seen in Figure 3, which all turned out to be temporary cache files. We do not have any attribution for the files given they are just cache files, we do not know who put them on the device.

Discussion

Our results from this investigation are inconclusive. Due to the difficulties we had mounting the drive image, we believe the main drive that held the evidence was corrupted somehow. We believe this may have happened either due to the device that read data from the

drive being faulty, or the drive itself being faulty. This issue of the corrupted drive greatly impacted our ability to perform a forensic investigation on it, and what data we did manage to pull from it ended up not being of value. There is data on this drive, but we don't know for sure if it pertains to the investigation or not.

Conclusion

In this investigation we first physically disassembled the suspect computer, taking out the battery to prevent it turning back on. Afterwards, we took out the internal storage drive and plugged it into the investigation computer. We then ran a script to hash the entire drive, take an image of the drive, then hash the image of the drive to ensure integrity of the image. We then re-imaged only the data portion of the drive due to issues from trying to do an investigation with the whole drive image, but got a result with 1 record missing. We then performed file carving on this image using PhotoRec, and extracted six total files. During analysis with grep and the strings command, these files just turned out to be temporary cache files. After this, we tried mounting the image, but due to issues with the image we always got an error trying. Then we ran our script again to re-image, ran the strings command on the image with grep to try and find Bible verses, but that didn't lead to any results. Our final findings from this resulted in the six temporary cache files.

References

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